

SALVADOR IBARRA LUNA

Senior SON & RAN Automation Architect | O-RAN/SMO Specialist | AI-Enabled Closed-Loop Network Operations

Location: Mexico (Remote)

Languages: English (Fluent), Spanish (Native)

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Portfolio: salvadoribarra.tech **GitHub:** github.com/josaiblu2

PROFESSIONAL SUMMARY

Senior **SON and RAN Automation Architect** with 23+ years designing, optimizing and automating 2G–5G radio networks for Tier-1 operators across multi-vendor ecosystems. Deep expertise in **closed-loop SON (cSON)** — automatic neighbor relations, PCI/RACH conflict resolution, capacity/coverage/interference optimization, load balancing and self-healing — delivered for **Verizon, AT&T, Sprint and Claro** on Ericsson, Nokia, Samsung and Huawei networks.

Extended this foundation into **O-RAN and SMO automation**: CU/DU integration, **O1 network management with NETCONF/YANG and 3GPP NRM**, CM/PM/FM data pipelines and rApp-aligned frameworks, including resolving **1,200+ KPI classification inconsistencies**. Now applying **AI-assisted engineering and human-in-the-loop closed-loop design** to the next generation of intelligent, autonomous network operations.

Recognized for leading multicultural teams, collaborating with R&D and vendors (Capgemini, Nokia, Ericsson, Samsung, Huawei), and delivering high-complexity First Field Applications and optimization programs. Committed to continuous learning, with **25 technical courses completed in 2026** across AI engineering, cloud-native systems and advanced 5G.

TECHNICAL SKILLS

SON / cSON Expertise

Closed-loop SON (cSON) • ANR / neighbor optimization • PCI & RACH conflict resolution • Capacity, coverage & interference optimization • Load balancing • Self-healing • DPO-L • RACO-L • CICO-L • ANE-L • DSH-L • DLB-L • VFBO • CAT-M1 • Multi-vendor integrations (Samsung–Nokia–ALU–Ericsson)

O-RAN & 5G Architecture

O-RAN O1 interface • SMO/SCAP workflows • CU/DU integration • 3GPP NRM • O-RAN Information Models • NETCONF/YANG • Sysrepo/Netopeer • Multivendor interworking (Capgemini, Benetel, Nokia, Ericsson, Samsung, Huawei)

Network Automation & Data Models

CM/PM/FM data pipelines • KPI dictionary creation • Data normalization • Consistency checks • Fault/Performance analytics • rApp functional workflows • Parameter audits • Scripting for PM analysis (Python basics, Bash)

AI-Ready Foundations

LLM-assisted troubleshooting (NETCONF, Linux) • AI-guided scripting for PM diff checks • AI-assisted research for FM/PM mapping • AI-supported ticket communication • AI-assisted software development (Claude Code, Cursor, MCP) • Human-in-the-loop closed-loop automation design

RAN Design & Optimization (2G–5G)

RF planning & optimization • PCI/PSC planning • Interference/capacity optimization • Neighbor relations • Field tests (drive/walk) • Log analysis • Physical/Logical design • KPI improvement (accessibility, retainability, mobility)

Tools

Atoll • Mentum Planet • Actix • ANA/Cisco-SON suite • Capesso • xAFP Optimi • Pathloss • MapInfo • Agilent/Nitro • R&S FSH3 • Anritsu Sitemaster

PROFESSIONAL EXPERIENCE

Independent RAN/SON Research & AI Automation Projects – Mexico (Remote)

Independent Projects | Jan 2026 – Present

- Conducted continuous technical research across SON, RAN, O-RAN, 5G and AI-enabled network automation, documented in **200+ technical posts in 2026** (177 focused on RAN/O-RAN/5G), with selected content published on salvadoribarra.tech.
- Architected and implemented an **AI-assisted closed-loop optimization framework** using AI agents under human-in-the-loop governance, incorporating experiment tracking, quality gates and KPI-driven performance evaluation, strengthening practical understanding of closed-loop principles applicable to intelligent network automation.

- Expanded hands-on software engineering capability through AI-assisted development of working applications (Python, FastAPI, TypeScript/Next.js, PostgreSQL, LLM APIs), complementing established SON/RAN architecture and automation expertise. Portfolio: github.com/josaiblu2.
 - Completed **25 technical courses (Jan-Jul 2026)** covering AI-assisted engineering, MCP, vector databases, Docker/Kubernetes, systems architecture, AI for 5G and telecom LLMs, Private 5G/NPN, 5G NTN and presales engineering.
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AWTG Ltd – UK (Remote)

R&D Network Operations Automation SME | Dec 2024 – Dec 2025

Architected and enhanced O-RAN management capabilities within AWTG's **SCAP SMO**, focusing on CM/PM/FM integration of Capgemini CU/DU and Benetel O-RU based on O1 specifications.

- Designed YANG-based data mappings, NETCONF operational workflows, and end-to-end CM/PM pipelines aligned with O-RAN and 3GPP NRM.
- Resolved **1,200+ KPI classification inconsistencies**, improving measurement accuracy and operational visibility.
- Contributed to the design of **rApps-aligned frameworks**: configuration audits, parameter consistency checks, fault correlation, and topology visualization.
- Improved FM/PM dictionary logic, enabling faster analytics, error detection, and automation-friendly data structures.
- Served as technical liaison with vendor engineering teams, ensuring seamless CU/DU/O-RU interworking.
- Used LLM-based tools to accelerate troubleshooting, test execution, scripting for PM diffs, and Netconf debugging.

Key technologies:

O-RAN O1 • NETCONF/YANG • 3GPP NRM • Sysrepo • KPI Dictionaries • CM/PM/FM Pipelines • rApps • SMO/SCAP • CU/DU Integration

HCL Technologies – Mexico/USA (Remote)

RAN Senior Technical Architect | 2021–2024

cSON consultant for Verizon across multiple markets (Virginia, New England, Pittsburgh, Houston, East Syracuse, Ohio).

- Supported and optimized cSON applications, driving performance improvements across LTE networks.
 - Designed **value-added analytical reports**, with ~55% authored personally, enabling detection of optimization opportunities and improving customer perception of service quality.
 - Coordinated features, validation cycles, and customer communications to achieve timely and high-quality deliveries.
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Iberica Industries / Tata Consultancy Services – Mexico/USA (Remote)

SON Senior RF Consultant – Customer Experience | 2019–2021

Supported First Field Application (FFA) of LTE Cisco-SON modules for Samsung USM platform and inter-vendor operability (Samsung–Nokia) for Verizon.

- Led FOA for Samsung USM, covering requirements analysis, test design, execution, defect management, and market adoption.
 - Delivered interoperability validations enabling automatic neighbor enhancements, PCI resolution, and capacity/interference improvements.
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Cisco Systems – Mexico / USA Remote

Radio Software Engineer – R&D | 2016–2019

Validated RF performance of new cSON features, vendor integrations, and new content for Tier-1 operators.

- Delivered complex **First Feature Application** for Nokia Airscale–FSM4, enabling multi-vendor SON operations (Nokia–ALU).
- Led Phase-2 Airscale program for Verizon, covering program management and product validation.
- Completed **CAT-M1 FFA for AT&T** in record time, collaborating onsite with developers in Israel.

- Identified critical multi-vendor issues for Sprint before mass deployment, avoiding major customer escalation.

Network Consulting Engineer – Customer Experience | 2014–2016

Led deployment, tuning, and support of Cisco-SON modules for Claro Central America (Ericsson & Huawei networks).

- Delivered full consulting scope early with zero complaints.
 - Developed high-impact analytical reports for detecting rehome errors, cross-feeders, PSC conflicts, etc.
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Ericsson (GSC) – Mexico

Senior Network Engineer – RAN Design | 2011–2014

Designed 2G/3G/4G networks for multiple North American and LatAm operators.

- Reassigned to AT&T USA critical projects due to performance.
 - Delivered coverage, quality, and capacity designs with high accuracy and fast turnaround.
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Nokia Siemens Networks – Mexico

Senior RF Optimization Engineer | 2010–2011

Responsible for RF optimization of the Nextel Matamoros, Tamaulipas region, supporting other markets on request.

- Improved retainability KPIs, achieving one of the lowest Drop Call Rates in market history.
 - Proposed capacity expansions, network modifications, and pre/post-launch assessments.
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Nextel – Mexico

iDEN RF Optimization Engineer | 2005–2010

Led performance monitoring, KPI improvement, neighbor planning, and optimization activities.

- Designed RF strategies that won the **national award for best-ranked market**.
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Earlier Experience (2000–2005)

Bordercomm – Project Manager (2005)

SCINTELSA – Microwave Consultant / Co-Owner (2003–2004)

SECOMSA – Microwave Engineering Manager & Interference Engineer (1999–2003)

EDUCATION

Communications and Electronics Engineering, Instituto Politécnico Nacional (1995–2000)

CERTIFICATIONS & TRAININGS

- 25 technical courses completed in 2026: AI-assisted engineering (Claude Code, Codex, Cursor, MCP, vector databases), Docker & Kubernetes, large-scale systems design, AI in 5G Networks & Telecom LLMs, Private 5G/NPN, 5G NTN, presales engineering
 - Swagger & OpenAPI Fundamentals (2025)
 - Generative AI & Midjourney (2025)
 - 5G KPIs & Optimization (2025)
 - 5G O-RAN Architecture & Use Cases (2024)
 - Fundamentals of DevOps, APIs, Microservices (2024)
 - Python Fundamentals (2024)
 - Leadership & Teams Management (2024)
 - Power of Storytelling (2024)
 - Professional Scrum Master I (2020)
 - Machine Learning: Executive Briefing – Google Cloud (2019)
 - SON v5 Certification – Cisco Intucell (2015)
 - Extensive additional RAN/LTE/5G trainings (Ericsson, Nokia, Cisco)
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TOOLS

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